Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 36)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 1 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 1 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

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0 2 . 1 Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

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0 2 . 2 Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 2 . 3 Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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5

0 3 A student compared the invertebrates in a stream above and below a sewage outfall using kick sampling.

0 3 . 1 Describe how kick sampling is carried out so that the samples from the two sites are comparable.

[3 marks]

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0 3 . 2 Give **three** limitations of kick sampling.

[3 marks]

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0 3 . 3 Explain how a Surber sampler improves on kick sampling.

[2 marks]

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0 3 . 4 Explain why the invertebrates found give a better indication of long-term sewage pollution than a single measurement of dissolved oxygen.

[2 marks]

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0 4 . 1 Explain **three** reasons why the development of new human activities has rarely been restricted by their environmental impacts.

[3 marks]

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0 4 . 2 Explain why sedentary rural communities often exploit resources sustainably.

[2 marks]

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0 4 . 3 Explain why urban populations are often unaware that the resources they use are being depleted.

[2 marks]

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0 4 . 4 Explain why the growth of affluence has increased per capita environmental impacts.

[3 marks]

10

0 5 . 1 Explain **three** features of natural systems that make them sustainable.

[3 marks]

0 5 . 2 Explain why the human use of mineral resources makes their sustainable exploitation much harder than the use of materials in a natural cycle.

[2 marks]

0 5 . 3 Explain **three** principles of the circular economy that should be applied when a new product is designed.

[3 marks]

0 5 . 4 Explain why an activity that is sustainable for a small population may be unsustainable for a large one. Use **one** example.

[3 marks]

0 5 . 5 Explain why urban populations are less likely than rural communities to notice that a resource is being over-exploited, and give **two** consequences of this.

[4 marks]

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15

0 6 . 1 Explain how GPS mapping, drone surveys and computer-based yield monitoring can reduce the environmental impacts of agriculture.

[3 marks]

0 6 . 2 Explain why productive agricultural technologies are less available in LEDCs.

[2 marks]

0 6 . 3 Explain how buying fairtrade food affects producers.

[1 mark]

0 6 . 4 Explain how nitrate from farmland can threaten human health.

[2 marks]

0 6 . 5 Give **two** agricultural sources of methane.

[2 marks]

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0 7 Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

Table 1

Species	Age at catchable size / years	Age at first breeding / years	Eggs per female per year	Maximum lifespan / years
Atlantic cod	2	4	Millions	25
Orange roughy	20	30	Thousands	150
Herring	2	2	Tens of thousands	20

0 7 . 1 Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

[4 marks]

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0 7 . 2 Explain why herring populations recover from overfishing much faster than cod.

[2 marks]

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0 7 . 3 Explain why the total mass of fish landed can stay the same while a population is being overfished.

[2 marks]

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0 7 . 4 Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species.

[2 marks]

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0 8 Figure 1 shows a simplified Antarctic food web.

Figure 1

Trophic level	Organisms
Producers	Phytoplankton, ice algae
Primary consumers	Zooplankton, krill
Secondary consumers	Small fish, squid, baleen whales, crabeater seals, penguins
Higher consumers	Toothfish, albatross, leopard seal, killer whale

0 8 . 1 Explain why the sea around Antarctica is so productive despite the cold.

[3 marks]

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0 8 . 2 Use Figure 1 to explain why fishing for krill and for Patagonian toothfish threatens other species.

[3 marks]

0 8 . 3 Explain the importance of Antarctica in controlling global natural cycles.

[3 marks]

0 8 . 4 Explain why the area of winter sea ice around Antarctica has increased in recent years even though air and sea are warming.

[1 mark]

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0 9 . 1 Explain the effects of deforestation on soil.

[4 marks]

0 9 . 2 Explain how deforestation affects hydrology and climate.

[4 marks]

0 9 . 3 Give **two** ecological impacts of deforestation other than the direct loss of species.

[2 marks]

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10.1 Evaluate the relative importance of ex-situ and in-situ methods in the conservation of endangered species.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **rewilding**.

[1 mark]

10

Write an essay on one of the following topics.

1 1 . 1 Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

OR

1 1 . 2 Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS